

STANDARD CLASS

Running Gemma 4 31B: locally and in the cloud

One 96 GB card locally, one H100 in the cloud. The model every FlatClaw tenant runs today. **Model:** Gemma 4 31B-IT · 31B dense · Apache 2.0 · 256K native · ~33 GB at FP8 + KV cache.

96 GB

VRAM, ONE CARD

FP8NATIVE ON BLACKWELL /
HOPPER**256K**

CONTEXT SERVED

8-12

CONCURRENT STREAMS

1

CARD, ONE PROCESS

LOCAL · BOUGHT ONCE

\$20,235

The one-card inference workstation as specced below; Newegg street prices on October 7, 2026.

CLOUD · HELD WARM

\$2,000/mo

Northflank nf-gpu-h100-80-1g, 1 × H100 80 GB; list price, 24/7, CPU and RAM bundled.

01 Local — The one-card inference workstation

One 96 GB card holds Gemma 4 31B at FP8 with the full 256K window, exactly as the hosted tenant runs it; the rest of the box is an ordinary high-end desktop platform. Every part is sold by Newegg today at the price shown.

PART	PICK · WHY IT'S HERE	STREET	BUY
GPU	NVIDIA RTX PRO 6000 Blackwell Max-Q, 96 GB 96 GB of GDDR7 holds the 31B at FP8 with the full 256K window on one card, exactly how the hosted tenant runs it. 300 W, a quiet blower, display outputs. The 600 W Workstation Edition is a few percent faster when Newegg has it in stock; today it does not. The passive Server Edition is wrong for a tower.	\$17,000 in stock	Newegg ↗
CPU	AMD Ryzen 9 9900X, 12 cores Feeds the card over PCIe 5.0 and runs the OS, SGLang, the gateway and ingestion. Twelve cores is plenty; the 9950X only pays off for CPU-heavy ingestion.	\$379 in stock · list \$403	Newegg ↗
MAINBOARD	ASUS ProArt X870E-Creator WiFi Two PCIe 5.0 x16 slots (x8/x8) so a second card can follow later, 10 GbE + 2.5 GbE, four M.2, USB4: workstation-grade AM5.	\$512 in stock	Newegg ↗
MEMORY	Corsair Vengeance 64 GB (2 × 32 GB) DDR5-6000 CL30 Two DIMMs keep AM5 at full speed; 64 GB carries the OS, Docker, the gateway and ingestion with headroom, because the model lives in VRAM. DDR5 has roughly quadrupled in price this year; 96 GB kits are not worth it for this box.	\$1,030 in stock · list \$1,069	Newegg ↗
POWER	Corsair HX1200i (2025), 1200 W, 80+ Platinum, ATX 3.1 A native 12V-2×6 lead for the Pro 6000 and wide margin over the ~500 W peak of card plus platform; the same unit carries the 600 W Workstation Edition if you swap in later.	\$288 in stock · list \$345	Newegg ↗
CPU COOLER	Corsair Nautilus II 360 RS, 360 mm AIO Holds the 120 W 9900X while the GPU dumps heat into the case; fits the North XL's top mount.	\$110 in stock	Newegg ↗
CASE	Fractal Design North XL, Mesh, charcoal and walnut A tower that looks like furniture: 413 mm of GPU room, a 360 mm top radiator, a mesh side feeding the blower. Mesh, not glass.	\$166 in stock · list \$195	Newegg ↗
STORAGE	Samsung 9100 PRO 4 TB, NVMe Gen5 OS plus the model library (~1 TB system, ~3 TB weights); the board takes Gen5 natively.	\$750 in stock · list \$1,360	Newegg ↗
All-in, as specced · 8 parts, 8 priced live from newegg.com on October 7, 2026		\$20,235	

Newegg street prices as read on October 7, 2026; the live table at flatclaw.org/builds refreshes hourly. Used-market and unlisted parts are marked. Verify at checkout.

02 Cloud — the same checkpoint, one service

NORTHFLANK · RECOMMENDED · nf-gpu-h100-80-1g

\$2,000/mo · \$2.74/hr

1 × H100 80 GB · 80 GB. 26 vCPU and 234 GB RAM bundled with the GPU; ~\$2,000/month held warm 24/7 including the bundled compute. Cold boot ~10 minutes: image pull, weights fetched from the weights-server pod over the internal network, SGLang load.

NORTHFLANK · nf-compute-400-16 (control plane)

\$144/mo

none. Portal plus one OpenClaw gateway per user: 4 vCPU / 16 GB carries two or three users at 0.6–1.2 GB each with room to spare. Every tenant has this service next to its GPU.

NORTHFLANK · nf-compute-50 + 200 GB volume

\$30/mo

none. weights-server: a CPU pod serving the tenancy's weights volume over HTTP, because GPU plans cannot mount the volume directly.

ALL-IN

About \$2,200 a month per tenant held warm: the GPU plan, the control plane and the weights-server. List prices; committed-use terms on any cloud cut the GPU line.

THE SAME SHAPE ELSEWHERE

- ▶ **Azure** · NC H100 v5 — AKS node pool or a single GPU VM in the customer's subscription; same containers, same weights-volume pattern (Blob + blobfuse).
- ▶ **AWS** · p5 / g6e class — EKS or a single GPU instance in an account the customer owns; IAM + KMS, private VPC egress only.
- ▶ **Google Cloud** · A3 (H100) — GKE with an A3 node or a single GPU VM; Workload Identity, VPC Service Controls.

NOTES

- ▶ Flat per-tenant rate, not per-token metering: the GPU line dominates and everything else is under \$200.
- ▶ Capacity is peak concurrent sessions, not head count: one H100 sustains 8–12 concurrent streaming chats at FP8 with first tokens in 1–2 s.

03 Run it

Serve

```
python -m sglang.launch_server --model-path google/gemma-4-31b-it --quantization fp8 --tp 1 --context-length 262144 --tool-call-parser gemma4 --reasoning-parser gemma4 --port 30000 --mem-fraction-static 0.90
```

- ▶ Peak draw is the card's 300 W plus about 200 W of platform: ~500 W, about 4 A at 120 V on an ordinary circuit. Quiet enough for a desk.
- ▶ Ubuntu 24.04 Server with NVIDIA R570+ / CUDA 12.8+ (sm_120). The card has display outputs for a monitor when you want one and runs headless otherwise. Serve from containers and hold driver updates before demo days.
- ▶ One SGLang process on the one card behind one OpenAI-compatible endpoint, the same launch line as the hosted tenant. The model never reloads.
- ▶ Run the OpenClaw gateway and the Portal on the same box for a fully local setup, or point a separate control machine at its endpoint; both work.